Apex Systems

Technical and Commercial Proposal

Response to RFP MRWB-2026-014: Customer Service and Billing Platform Modernisation

Issued by: Meridian Regional Water Board (fictional)

Submitted: 2026-10-20

Item	Detail
Bidder	Apex Systems Pvt. Ltd., Pune, India
Proposed solution	Apex UtilityCloud: cloud-native billing and customer service platform
Contract duration	40 weeks implementation plus 3-year subscription
Total 3-year cost	INR 14.21 Cr (fixed price implementation), excludes GST
Certifications	ISO 27001:2022, SOC 2 Type II
Bid validity	150 days from submission

This is a fictional proposal created for a classroom exercise. All organisations, people and figures are invented.

Table of Contents

1. Executive Summary and Understanding of the Requirement

2. Proposed Solution and Architecture

3. Integration Approach

4. Implementation Approach

5. Timeline, Team Structure and Milestones

6. Commercial Proposal and Price Table

7. Security, Compliance and Risk Controls

8. Support Model and Service Level Agreement

9. Relevant Experience and References

10. Assumptions and Exclusions

11. Appendices

1. Executive Summary and Understanding of the Requirement

Apex Systems proposes a cloud-native Customer Service and Billing Platform for the Meridian Regional Water Board that replaces the 14-year-old legacy billing system and three disconnected customer service tools. Our solution is designed for 1.2 million service connections today and scales to 3 million without re-architecture.

Apex has deep engineering capability and a mature security programme. We hold ISO 27001:2022 and SOC 2 Type II attestations, and our platform has been independently penetration tested every six months since 2021. We believe security and resilience are non-negotiable for critical utility infrastructure.

What we understood from the RFP

The Board needs to consolidate billing, metering data, customer self-service and field-service workflows; integrate with SAP S/4HANA finance and the existing AMI meter data management system (MDM); comply with data protection obligations under the Digital Personal Data Protection (DPDP) Act; and reduce billing exceptions, currently around 4.5 percent of monthly bills.

From the clarification session and the data room we also noted four constraints that shaped our design: (a) 14 years of customer history, roughly 38 million billing records, must be retained and searchable; (b) the Board operates 11 zone offices with uneven connectivity, so the agent desktop must degrade gracefully; (c) tariff policy changes twice a year and must be configurable by Board staff without vendor involvement; and (d) the billing run window is 8 hours overnight and must not extend.

Why Apex

- Security first: ISO 27001:2022 and SOC 2 Type II, with VAPT every six months and a 1-hour breach notification commitment.
- Engineering depth: 62 platform engineers, a dedicated utilities product team and a 99.95 percent measured availability over the last 24 months.
- Clear commercials: fixed-price implementation, published rate card, no open-ended time-and-materials exposure for the Board.
- Proven utility delivery: 6 utility billing implementations in 8 years, including two water utilities.

Value proposition summary

Board objective	How Apex addresses it	Measure
Reduce billing exceptions	Pre-bill validation engine with 140 configurable rules and exception work queues	Exceptions from 4.5 percent to below 1.5 percent within 6 months of go-live
Faster bill cycle	Parallelised billing engine, 12 worker pods per zone group	1.2 million accounts billed in under 3 hours
Digital self-service	Portal and mobile app in 4 languages with UPI payments	Target 55 percent digital payment share by month 12
Regulatory compliance	Consent management, India data residency, 7-year audit logs	Zero audit findings at first annual review (target)

2. Proposed Solution and Architecture

The platform uses a microservices architecture deployed on Kubernetes across two cloud availability zones in an Indian region, with active-active failover and a recovery point objective (RPO) of 5 minutes and recovery time objective (RTO) of 30 minutes. Each domain service owns its data store; services communicate through Apache Kafka events and a REST API gateway.

- Billing engine: rule-based tariff configuration supporting slab, seasonal and subsidy tariffs without code changes.
- Integration layer: event streaming with Apache Kafka plus a REST API gateway; 42 pre-built connectors including SAP S/4HANA (IDoc and OData) and MDM via MultiSpeak.
- Customer portal and mobile app: bill view, UPI payment, complaint tracking, outage notifications in English, Hindi and two regional languages.
- Performance: load tested to 2,500 concurrent portal users and a full billing cycle of 1.2 million accounts in under 3 hours.
- Data migration: automated reconciliation with 100 percent record-count and 99.98 percent value matching targets before cutover.

Logical architecture layers

Layer	Components	Technology and design notes
Channels	Customer web portal, mobile app (Android and iOS), agent desktop, IVR gateway, WhatsApp notifications	React front end, server-side rendering for portal pages, offline-tolerant agent desktop with local queueing for 11 zone offices
API and integration	API gateway, event bus, 42 connectors, orchestration service	Kong-class gateway with OAuth 2.0 and mTLS; Kafka with 3x replication, 30-day topic retention
Core services	Billing, rating, payments, CRM and case management, field-service dispatch, notifications	Java 21 and Kotlin services on Kubernetes; stateless pods, horizontal autoscaling on queue depth
Data	Operational store, analytics store, document store, search index	PostgreSQL clusters with synchronous replication inside a zone, asynchronous across zones; object storage for bill PDFs
Platform and security	Identity, secrets, key management, observability, CI/CD	Customer-managed keys in HSM-backed KMS, OpenTelemetry tracing, GitOps deployment with signed images

Resilience and scalability

Every stateful tier is replicated across two availability zones. Quarterly game-day exercises simulate zone loss, database failover and message-bus degradation. The billing engine scales from 12 to 48 worker pods during the monthly run and releases capacity afterwards. Capacity headroom is set at 2.5x the current peak, and the architecture was validated at 3 million connections in a synthetic test in August 2026 (billing cycle of 3 million accounts in 6 hours 40 minutes).

Configurability for Board staff

Tariff tables, rebate rules, late-fee policies and bill templates are maintained in an administration console with maker-checker approval and effective-dated versions. Board staff can simulate a tariff change against a sample of 50,000 historical accounts before publishing, and every change is written to the immutable audit log. Typical tariff revisions are completed in 2 working days without vendor involvement.

Component inventory

Component	Responsibility	Scaling and data notes
Account service	Customer, premise, service point and contract records; hierarchy for commercial accounts	Read replicas per zone; 38 million historical rows partitioned by billing year
Rating service	Applies slab, seasonal, subsidy and sewerage-linked tariffs; effective-dated tariff versions	Stateless; 12 to 48 pods; rating throughput 2,100 accounts per second per 12 pods
Bill presentation service	Generates bill PDFs in 4 languages with QR code for UPI payment	Object storage with 7-year retention; PDF render 400 bills per second
Payments service	Online, counter and agent collections; allocation to bills; refunds and reversals	Idempotent API; daily settlement file reconciliation with banks
Case management	Complaints, service requests, field work orders, SLA timers	Event driven; escalation rules configured by Board staff
Notification service	SMS, email, WhatsApp and push messages in preferred language	Template library of 160 messages; quiet hours per customer
Analytics store	Operational dashboards, revenue assurance reports, exception analytics	Columnar store refreshed every 5 minutes from the event stream

Non-functional targets

Attribute	Target	How verified
Portal page response (95th percentile)	Under 1.2 seconds at 2,500 concurrent users	Performance test in phase 4, repeated after each release
Billing run	1.2 million accounts in under 3 hours; 3 million in under 7 hours	Synthetic run at 3 million accounts, August 2026
Availability	99.95 percent monthly	Gateway probes every 30 seconds; monthly SLA report
Recovery	RPO 5 minutes, RTO 30 minutes	Twice-yearly disaster recovery test with the Board present
Accessibility	WCAG 2.1 level AA for portal	Independent accessibility audit before UAT

3. Integration Approach

Integration is the highest technical risk in this programme, so we designed it early and in detail. The following integration specification reflects the interface documents shared in the RFP data room.

Interface	Direction	Pattern and protocol	Volume and timing
SAP S/4HANA finance (GL, AR)	Outbound and inbound	IDoc over RFC for postings; OData for balance enquiry; nightly reconciliation file as fallback	About 1.2 million posting lines per cycle; near-real-time for payments
AMI meter data management (MDM)	Inbound	MultiSpeak 5 web services for reads; Kafka ingestion for interval data; validation, estimation and editing rules	Up to 1.4 million reads per day; 15-minute interval data for 90,000 commercial meters
UPI and payment gateways	Bidirectional	REST and webhooks with idempotency keys; signed callbacks	Peak 120 transactions per second
SMS, email and WhatsApp gateways	Outbound	REST with delivery receipts; templated multilingual messages	Up to 2.4 million messages per cycle

Interface	Direction	Pattern and protocol	Volume and timing
Legacy billing (migration only)	Inbound	Read-only extract through staged flat files and database link; checksum-controlled	38 million historical billing records, three mock migrations

Integration governance

Each interface has a named owner on both sides, a signed interface control document by week 8, a contract-test suite that runs on every build, and a certification gate in week 26. A virtualised SAP and MDM simulator allows development to proceed before the Board's test environments are ready, which protects the schedule if environment delivery slips by up to three weeks.

Error handling

All integrations use at-least-once delivery with idempotent consumers, dead-letter queues and an operations console where support staff can inspect and replay failed messages. Alerts fire when any interface backlog exceeds 15 minutes.

4. Implementation Approach

Delivery follows a phased hybrid approach: design waterfall for regulated billing rules, then agile two-week sprints for portal and integrations. Each phase ends with a documented exit review attended by the Board's PMO.

Phase	Key activities	Exit criteria
1. Discovery and design	Process workshops in all 11 zones, tariff catalogue, interface control documents, data profiling of legacy records	Signed solution design, tariff catalogue and migration strategy
2. Build and integration	13 sprints; core billing, portal, mobile, SAP and MDM integrations; security hardening	SAP and MDM integrations certified; 92 percent of user stories accepted
3. Data migration	Three mock migrations with automated reconciliation; cleansing of duplicate and orphan accounts	Mock 3 reconciles at 99.98 percent value match
4. Testing	System integration test, UAT in 3 zones, performance test at 2,500 users, VAPT, disaster recovery test	UAT sign-off, no open Severity 1 or 2 defects, VAPT findings closed
5. Go-live and hypercare	Zone-wise cutover over three weekends, 4 weeks hypercare with on-site support	Stable billing cycle, handover to run support

Change management and training

A change network of 22 Board champions (two per zone office plus HQ) is trained in weeks 20-26. Training covers 30 train-the-trainer sessions, role-based e-learning modules and a sandbox environment. Super-users receive 5 days of training; agents receive 2 days. Training completion of 95 percent is an exit criterion for UAT.

Quality assurance

A test pyramid is enforced in the pipeline: at least 80 percent unit test coverage on billing services, 1,400 automated regression scenarios and a nightly performance smoke test. Defect severity definitions follow the SLA in Section 8.

Data migration plan

Data migration is run as a controlled sub-project with its own lead and its own exit gates. Legacy data is extracted read-only into a staging area, profiled, cleansed, transformed into the new account model and loaded. Every load produces a reconciliation pack that the Board's finance team signs.

Step	Weeks	Activity	Control
Profile	3-8	Profile 38 million billing records, 1.2 million accounts, 2.9 million payment histories; identify duplicates, orphans, invalid tariffs	Data quality scorecard reviewed weekly
Cleanse	9-20	Two cleansing cycles with Board data stewards; merge duplicate customers; fix invalid meter links	Steward sign-off on every merge rule

Step	Weeks	Activity	Control
Mock 1	20-23	Full load of accounts and 24 months of billing history; measure load time and error rate	Record count 100 percent; value match target 99.9 percent
Mock 2	25-28	Full history load, open balances and deposits; parallel bill calculation for 50,000 accounts	Parallel bill variance under 0.05 percent
Mock 3	30-32	Dress rehearsal of cutover with timing; rollback tested	Value match 99.98 percent; open balances exact
Cutover	37	Final delta load over a weekend freeze; rollback decision point at hour 20	Go or no-go signed by Board sponsor

Test strategy

Test level	Scope	Entry / exit	Volume
Unit and component	Services and rules, run in CI on every commit	Coverage at least 80 percent on billing services	About 9,000 automated tests
Contract tests	Every integration interface against SAP and MDM simulators	All contracts green before certification gate	320 contract cases
System integration	End-to-end meter-to-cash scenarios	No open Severity 1 or 2 defects	620 scenarios
Performance	Billing run, portal load, API spikes, failover	Targets in Section 2 met on two consecutive runs	6 scripted load profiles
Security	SAST, DAST, dependency scan, independent VAPT	No open critical or high findings	Per release and six-monthly
User acceptance	Board users in 3 pilot zones, 4 weeks	Signed UAT certificate, 95 percent training completion	480 business scripts

5. Timeline, Team Structure and Milestones

Total duration is 40 weeks. The plan below shows phase windows and the payment-linked milestones.

Phase	Weeks	Key milestone
Discovery and design	1-8	Approved solution design and tariff catalogue
Build and integration	9-26	SAP and MDM integrations certified
Data migration rehearsals	20-32	Three mock migrations passed
UAT and performance testing	27-36	UAT sign-off
Go-live and hypercare	37-40	Go-live plus 4 weeks hypercare

Team structure

Team: 1 delivery director, 1 project manager, 2 solution architects, 8 engineers, 2 QA leads and 1 security architect (14 FTE at peak, 11 FTE average). A risk register is reviewed weekly with the Board's PMO and a steering committee meets monthly.

Role	FTE	Responsibility	Location
Delivery director	0.3	Executive sponsor, escalation, steering committee	Pune
Project manager	1.0	Plan, risks, dependencies, reporting	On site, Meridian HQ
Solution architects	2.0	Overall design, integration design, non-functional requirements	1 on site, 1 Pune
Senior engineers	8.0	Billing, portal, integration, data migration	Pune with rotating on-site presence
QA leads	2.0	Test strategy, automation, performance testing	Pune
Security architect	1.0	Threat modelling, VAPT coordination, compliance evidence	Pune

Governance

Weekly progress review with the Board's PMO, fortnightly sprint demonstrations, monthly steering committee, and a formal change control board. Status reports include earned-value indicators, defect trends and a RAG-rated risk list.

Top delivery risks

Risk	Likelihood	Mitigation
SAP or MDM test environments late	Medium	Virtualised simulators; environment booking in week 1; 3-week schedule buffer
Poor quality of legacy data	High	Profiling in weeks 3-6; two cleansing cycles; reconciliation gates
Tariff rules not frozen on time	Medium	Tariff freeze at design sign-off; change control after week 8

Responsibility matrix (RACI)

Activity	Apex delivery	Apex architect	Board PMO	Board business owner
Project plan and reporting	A / R	C	R	I
Tariff catalogue sign-off	C	C	I	A / R
Interface control documents	A	R	C	C
SAP and MDM environments	C	C	A / R	I
Data cleansing decisions	C	C	I	A / R
UAT execution	C	I	A	R
Go-live decision	R	C	C	A

R = responsible, A = accountable, C = consulted, I = informed.

Sample fortnightly status report (illustrative extract)

Item	Status	Comment
Schedule	Green	Sprint 7 of 13 complete; velocity 46 points against plan of 44
Scope	Green	2 change requests approved, both within contingency of 120 person-days
Risk	Amber	SAP test client refresh delayed 4 days; simulator in use; escalated to Board PMO
Quality	Green	Open defects: 0 Severity 1, 3 Severity 2, 41 Severity 3; automation pass rate 98.6 percent
Budget	Green	Earned value index 1.02; 41 percent of implementation fee invoiced against 40 percent progress

6. Commercial Proposal and Price Table

Item	Price (INR)	Notes
Implementation services (fixed price)	6,80,00,000	Milestone-based payments
Platform subscription, Year 1	2,60,00,000	Up to 1.5 million connections
Platform subscription, Year 2	2,60,00,000	Price held flat
Platform subscription, Year 3	2,73,00,000	5 percent uplift
Data migration and training	48,00,000	Includes 30 train-the-trainer sessions
Total 3-year cost	14,21,00,000	Excludes GST

Assumptions: the Board provides SAP integration test environments by week 9; tariff rules are frozen at design sign-off; change requests are charged at INR 18,500 per person-day. Payment terms are 20 percent on design sign-off, 30 percent on UAT, 40 percent on go-live and 10 percent after hypercare.

Payment schedule for implementation services

Milestone	Percent	Amount (INR)
Design sign-off (week 8)	20	1,36,00,000

Milestone	Percent	Amount (INR)
UAT sign-off (week 36)	30	2,04,00,000
Go-live (week 37)	40	2,72,00,000
After hypercare (week 40)	10	68,00,000

Pricing basis and rate card

Our price is a fixed price for the scope described in this document. Subscription fees cover hosting in two availability zones, 24x7 support, platform upgrades and security monitoring. Fees for connections above 1.5 million are INR 14 per connection per month. Year-3 subscription includes a 5 percent uplift; no other escalation applies in the 3-year term.

Role (for change requests)	Rate per person-day (INR)
Solution architect	24,000
Senior engineer	18,500
QA engineer	14,000
Blended rate for change requests	18,500

We acknowledge that this is the highest-priced proposal in the market range we expect. The premium reflects dual-zone active-active hosting, 24x7 SIEM monitoring, six-monthly independent penetration testing and a larger delivery team (14 FTE at peak). We are open to discussing scope-based options, for example single-zone hosting for non-production environments, to reduce cost.

7. Security, Compliance and Risk Controls

- Certifications: ISO 27001:2022 (certificate valid to 2028) and SOC 2 Type II report available under NDA.
- Encryption: AES-256 at rest with customer-managed keys; TLS 1.3 in transit.
- Access: MFA for all users, role-based access control and privileged access management with session recording.
- Monitoring: 24x7 SIEM, immutable audit logs retained for 7 years, quarterly access reviews.
- Privacy: DPDP Act consent management, data residency in India, data protection impact assessment in design phase.
- Testing: independent VAPT before go-live and every six months; incident response plan with 1-hour notification.

Security governance

Security is led by a named security architect on the project and by Apex's Chief Information Security Officer at corporate level. Threat modelling (STRIDE) is performed for every major service during design, and secure coding standards aligned to OWASP ASVS Level 2 are enforced through static analysis, dependency scanning and signed container images. Production access is just-in-time, approved and recorded.

Risk register (security and operational)

Risk	Impact	Control	Residual
Unauthorised access to customer data	High	MFA, RBAC, PAM with session recording, quarterly access review	Low
Ransomware or destructive attack	High	Immutable backups every 15 minutes, offline copy daily, restore test each quarter	Low
Zone outage in cloud region	High	Active-active across two zones; RPO 5 minutes, RTO 30 minutes; game days	Low
Payment fraud via portal	Medium	Device fingerprinting, velocity limits, tokenised payments, PCI DSS scope reduction	Low

Risk	Impact	Control	Residual
Privacy breach under DPDP Act	High	Consent ledger, field-level encryption, DPIA, 1-hour internal escalation, 72-hour regulator notice process	Low
Supply-chain compromise	Medium	SBOM per release, signed images, vendor assessments, dependency pinning	Medium

Business continuity

The disaster recovery plan is tested twice a year with the Board's participation. Backups are encrypted, replicated across zones and retained for 35 days online and 7 years archived. A documented incident response plan commits Apex to notify the Board within 1 hour of confirmed personal-data compromise and to provide a root-cause report within 5 working days.

Risk	Impact	Control	Residual
Insider misuse of privileged access	High	Just-in-time access, approvals, session recording, monthly privileged access review	Low
Misconfiguration of cloud storage	High	Policy-as-code scanning, deny-by-default network rules, weekly posture report	Low
Denial of service on portal	Medium	Edge protection, rate limiting, autoscaling, tested at 5x peak	Low
Loss of key person in project	Medium	Named deputies, documented design, cross-training in first 12 weeks	Medium
Integration defects found late	Medium	Contract tests on every build, certification gate in week 26	Low
Training insufficient for zone staff	Medium	Sandbox practice, 22 champions, completion gate at 95 percent	Low

Compliance mapping

Obligation	Platform control	Evidence provided to Board
DPDP Act consent and purpose limitation	Consent ledger with purpose tags; field-level masking for agents	Consent report, DPIA, sample masked views
Data residency in India	All stores and backups in Indian regions; no cross-border replication	Hosting attestation and architecture diagram
Audit and retention	Immutable logs, 7-year retention, tamper-evident hashing	Quarterly audit log extracts
Financial reconciliation	Daily payment settlement matching, exception queues	Daily reconciliation report to Board finance
Vendor security assurance	ISO 27001:2022 certificate, SOC 2 Type II report	Certificate and report summary under NDA

8. Support Model and Service Level Agreement

Support is provided 24x7 for Priority 1 incidents with a 30-minute response time and 4-hour resolution target; Priority 2 within 2 hours. A named service manager runs monthly service reviews.

Priority	Definition	Response	Resolution target	Update frequency
P1 Critical	Billing or payments unavailable, data breach	30 minutes	4 hours	Every 30 minutes
P2 High	Major function degraded, workaround exists	2 hours	8 hours	Every 2 hours
P3 Medium	Minor function impaired	1 business day	5 business days	Daily
P4 Low	Question or cosmetic issue	2 business days	Next release	Weekly

Platform availability commitment is 99.95 percent per month, measured at the API gateway, excluding agreed maintenance windows (maximum 4 hours per month, Sundays 01:00-05:00). Service credits are 5 percent of

the monthly subscription for each 0.1 percent below target, capped at 25 percent.

Support organisation

Tier 1 is a 24x7 service desk in Pune; Tier 2 is the application support team (9 engineers across three shifts); Tier 3 is the product engineering team with on-call rotation. During hypercare, 4 engineers are on site at Meridian HQ and zone offices. A named service manager chairs monthly service reviews and a quarterly roadmap review.

Service level indicators and reporting

Indicator	Target	Measurement
Platform availability	99.95 percent per month	API gateway probes every 30 seconds
Billing run completion	Within 8-hour overnight window, 100 percent of cycles	Run log reviewed by service manager
Payment posting latency	95 percent within 2 minutes of gateway confirmation	Payments service metrics
P1 response / resolution	30 minutes / 4 hours	Ticket timestamps
P2 response / resolution	2 hours / 8 hours	Ticket timestamps
Backup success	100 percent of scheduled backups; restore test quarterly	Backup system report
Customer satisfaction on service desk	At least 4.3 out of 5	Post-ticket survey

A monthly service report with availability, incident trends, problem records, capacity forecast and security events is delivered by working day 5. Major incidents receive a written post-incident review within 5 working days. The service desk escalation path runs from the service desk lead to the service manager, delivery director and, for sustained P1 incidents beyond 8 hours, the Apex managing director.

9. Relevant Experience and References

Apex has completed 6 utility billing implementations in the last 8 years, including two water utilities. Our utilities practice has 85 staff and a retention rate of 91 percent over the last three years.

Reference client (fictional)	Scope	Year
Sundara City Water Corporation	Billing and CRM for 800,000 connections	2024
Eastern Grid Power Distribution	Billing platform and SAP integration	2022

Reference outcomes

Reference	Outcome	Contact
Sundara City Water Corporation	On-time go-live in 38 weeks; billing exceptions reduced from 5.1 to 1.2 percent; 99.96 percent availability in first year	Chief Information Officer; consent given for call
Eastern Grid Power Distribution	SAP S/4HANA integration live in 31 weeks; bill run time cut from 11 hours to 2 hours 40 minutes	Head of Revenue Systems; consent given for call

Both references have agreed to reference calls and a site visit. The Sundara programme is the closest analogue to this contract: a water utility with a customer base of 800,000 connections, SAP integration and a portal in three languages. Lessons carried into this proposal include starting tariff workshops in week 1, freezing tariff scope at design sign-off and rehearsing cutover three times.

Capability area	Apex evidence	Count
Utility billing implementations	Water, electricity and distribution clients	6 in 8 years
SAP S/4HANA integrations	Finance postings with IDoc and OData	4

Capability area	Apex evidence	Count
Platform security audits	ISO 27001:2022, SOC 2 Type II, VAPT every six months	Continuous since 2021
Cutovers of more than 500,000 accounts	Weekend freeze with rollback tested	3

10. Assumptions and Exclusions

Assumptions

- The Board provides SAP S/4HANA and MDM test environments by week 9, with technical contacts available within 2 working days.
- Tariff rules and billing policies are frozen at design sign-off (week 8). Later changes follow change control.
- The Board nominates a single product owner with decision authority, and subject-matter experts for 2 days per week during design and UAT.
- Legacy data extracts are provided in agreed formats by week 12; data quality issues beyond 2 cleansing cycles are a change request.
- Cloud hosting is in an Indian region of Apex's chosen provider; the Board has no preference for a specific provider.

Exclusions

- Replacement or upgrade of the Board's SAP S/4HANA and MDM systems.
- Meter hardware, network connectivity upgrades at zone offices and end-user devices.
- Paper bill printing and postal services; SMS, email and WhatsApp message charges (billed at actuals by the gateway providers).
- GST and any statutory levies, which are payable in addition to the quoted prices.
- Digitisation of paper records older than the 14-year legacy system history.

11. Appendices

Appendix A: Key staff summaries

Name (fictional)	Role	Experience
Rohan Deshpande	Delivery director	22 years in enterprise delivery; led Sundara City Water programme (800,000 connections); PMP, ITIL 4
Meera Iyer	Project manager	15 years; delivered 4 utility billing programmes; PMP and PRINCE2 Practitioner
Dr. Kabir Anand	Lead solution architect	19 years; designed Kafka-based billing platforms; TOGAF 9 certified; ex-SAP integration lead
Latha Ramaswamy	Security architect	14 years; CISSP and CISM; led ISO 27001 and SOC 2 Type II audits for Apex
Imran Sheikh	QA lead	12 years; performance engineering of 3 million account billing runs; ISTQB Advanced

Appendix B: Glossary

Term	Meaning
AMI	Advanced metering infrastructure, the smart meter network
MDM	Meter data management system holding validated meter reads
RPO / RTO	Recovery point objective (data loss tolerance) and recovery time objective (restoration time)
VAPT	Vulnerability assessment and penetration testing

Term	Meaning
DPDP Act	Digital Personal Data Protection Act, India
SIEM	Security information and event management
UAT	User acceptance testing by the Board

Appendix C: Compliance statement

Apex confirms that this proposal complies with all mandatory requirements in RFP MRWB-2026-014 and that no deviations are taken. Certificates and the SOC 2 Type II report summary will be shared with the Board's evaluation committee on request under NDA.

Appendix D: Training plan summary

Audience	People	Duration	Format
Board super-users and champions	22	5 days	Classroom plus sandbox exercises
Contact-centre and counter agents	320	2 days	Train-the-trainer in 30 sessions
Finance and revenue assurance	45	2 days	Workshops on reconciliation and adjustments
Field supervisors	60	1 day	Mobile work-order training
IT operations	18	5 days	Run, monitor, deploy and recover the platform

Appendix E: Exit and transition

At contract end or termination Apex provides all Board data in documented open formats within 30 days, retains no copy after written confirmation, and provides up to 12 weeks of transition support at the rate card. Source configuration (tariffs, templates, workflows) is exported in machine-readable form.

Appendix F: Clarifications offered to the Board

Topic	Apex position
Hosting alternatives	Single-zone production hosting is available at INR 28,00,000 lower per year, with RTO of 4 hours; not recommended
Optional scope	Additional regional language, INR 9,50,000 per language; extra mobile app features priced per sprint
Contract term	Subscription may be extended for 2 further years at no more than 5 percent annual uplift
Source code escrow	Available with an independent escrow agent at the Board's request
Reference visits	Apex will arrange site visits to Sundara City Water Corporation within 3 weeks of request

Apex is available to meet the evaluation committee for a demonstration of the billing engine, the tariff simulator and the operations console at the Board's convenience.

End of proposal. This is a fictional proposal created for a classroom exercise; all organisations, people and figures are invented.